

MECG15603/MCCG15603/UECS3153

ASSIGNMENT 2

COURSE NAME: STATISTICAL LEARNING

PROGRAMME: DSC, DTC, SE

Instructions

1. In this assignment, a group with 3 to 5 members will be formed to write **computer program(s) in R programming language (10%)**, **a report (20%)** and to present your results in **an oral presentation (10%)** with at least $N - 1$ supervised learning models and at least $\text{ceiling}(N/2) - 1$ unsupervised learning models (from Week 2–Week 4) when the number of members is N . The following are the course learning outcomes (CLO) relevant to the assessment:
 - CLO2: Compare statistical models through supervised learning for prediction and estimation Report
 - CLO4: Demonstrate results from optimised supervised and unsupervised learning models. R script, Oral Presentation
2. To prevent you from generating programs using AI, you should only use basic R functionalities based on `data.frame` and should prevent using advanced R frameworks such as `dplyr`, `tidyverse`, `caret`, etc. If you use any of them, you need to write an equivalent form in basic R in the comments. Otherwise, marks will be deducted.
3. The group leader is responsible for combining all contributions from the group members and write down the contributions of each member in quantitative measurements. **Any member who contributes nothing to the assignment will only receive 60% of the group assignment marks.**
4. The **deadline of the submission** is **9:30pm Tuesday Week 12** (1st September 2026). Both the R script (readable by the notepad editor) and the assignment report (in PDF or in Word document format) can be submitted through email (liewhh@utar.edu.my) or MS Teams Chat (to Liew How Hui) by the group leader.
5. In the case of **late submission** for the report and program script, 10% of the marks will be deducted if the work is up to one day late (24 hours) and additional 10% of the maximum marks for each of the subsequent days.
6. **Plagiarism is not allowed.** If the works are found to be plagiarised literally from the Internet, no marks will be awarded and the incident will be reported to the university for further action.

Part 1: Programming Lab (10%)

1. Pick **one** dataset from the following list of case studies and perform **unsupervised** and **supervised** learning on the selected dataset for a particular case study:
 - Gallstone Dataset (<https://archive.ics.uci.edu/dataset/1150/gallstone-1>): The dataset is in Excel format and being zip twice. In this case study, you need to implement a program which can work with the original data in Excel format to learn the significant features which contributes to gallstone. The more we learn how to predict different illnesses can allow us to build a better healthcare system that provides early detection of illnesses and hopefully supports early prevention. The relevant Sustainable Development Goal for this case study is SDG 3 (Good health and well-being — Reduce mortality from non-communicable diseases).
 - Website Phishing Data Set (<https://archive.ics.uci.edu/dataset/327/phishing+websites>) The dataset is in ARFF format (https://waikato.github.io/weka-wiki/formats_and_processing/arff/) in a zip file. In this case study, you need to implement a program which can work with the original data in ARFF format to learn the significant features which are relevant to phishing using unsupervised and supervised models. The relevant Sustainable Development Goal for this case study is SDG 16 (Peace, justice, and strong institutions — combat organised crime).
 - Spambase Dataset (<https://archive.ics.uci.edu/dataset/94/spambase>) The “spam” concept is diverse, it includes advertisements for products/web sites, make money fast schemes, chain letters, pornography. In this case study, you need to analyse the data and build a model which can detect whether a given email is spam or not based on the features. We can allow around 7% misclassification error but false positives (marking good mail as spam) are very undesirable. The relevant Sustainable Development Goal for this case study is SDG 16 (Peace, justice, and strong institutions — combat organised crime).
2. Write a (nicely structured) R programming code or R codes with the use of appropriate libraries which analyse the selected **raw dataset** and are directly executable by the R interpreter (<https://cran.r-project.org/>).
3. The programming code can only use free and legal statistical software such as R, RStudio or MCRAN. The code should have reasonable dependencies and is cross-platform, i.e. the program can run on Microsoft Windows, GNU/Linux platform, MacOS, etc.
4. Marks will be **deducted** if data processing taught in the practical are not used but the sophisticated techniques from the Internet or AI are used (such as **dplyr**, **tidyverse**, **caret**, etc.) without proper documentation/rationale in the programming code.
5. The programming code(s) need to perform feature analysis, feature-response analysis, at least $\text{ceiling}(N/2) - 1$ pattern analysis (dimension and/or clustering) and then **compare** at least $N - 1$ supervised learning models for prediction and estimation. Here N is the number of members in a group. The results should be properly organised in the assignment report.

Part 2: Group Assignment Report (20%)

1. You should document the following analysis results derived from the programming lab in Part 1 into your assignment report.
 - the results of unsupervised learning with appropriate feature analysis, feature-response analysis and pattern discovery of the data dimension and possible clustering patterns.
 - the results of supervised learning by comparing at least $N - 1$ number of supervised learning models for their performance. Here N is the number of members in an assignment group.
2. The assignment report should be written in a proper report format with the following components:
 - A page cover with a title, information on members, etc. as well as a table of contents;
 - An introduction is comprehensive (documenting the background of the data) and has appropriate academic references and problem solving objective(s) for the case study;
 - A chapter with appropriate unsupervised learning with feature analysis and feature-response analysis using methods from exploratory data analysis, appropriate pattern analysis including dimension reduction (and clustering pattern analysis if there more than 4 members) based on programming lab results;
 - A chapter with the training of at least $N - 1$ supervised learning models and their performance analysis summarised in a table at the end of the chapter. The performance and structure of trained models should be properly studied;
 - A conclusion.

Note that a good report should have appropriate theories and academic references for appropriate unsupervised learning and supervised learning as well as logical and appropriate presentations (using either tables or statistical diagrams) for the demonstration of results.

Part 3: Presentation (10%)

1. Prepare presentation slides which summarises the group assignment report and the programming scripts.
2. An oral presentation which involves all members or selected member(s) are allowed.
3. The oral presentation should cover the following aspects:
 - A good introduction of the case study with relevant references as well as the sourcing of data, data loading, data transformation (for unstructured data), type checking (for structured data), cleaning and proper univariate analysis.
 - A good illustration of the discoveries of the statistical patterns (statistical distribution or clustering patterns) of each feature with respect to the response as well as the combinations of features with respect to the response with the use of unsupervised learning methods.
 - A good illustration of the trained supervised learning models. If the trained model meets the prediction requirement, explain how the model could be deployed and updated; OR
If the trained model does not meet the prediction requirement, explain the possible reasons from statistical and/or mathematical perspective.
 - The presentation is concluded with a comparison table for comparing the performance of the trained supervised models and how the objectives have been met as well possible future research work.
 - The presentation is well-timed (less than 10 minutes but more than 6 minutes, heavy marks may be deducted for the presentation which is over-time) and well-prepared.